

ORDER NO. 98-039

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF GUSTINE
WASTEWATER TREATMENT FACILITY
AND
SUBSURFACE DRAINAGE SYSTEM
MERCED COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Gustine (hereafter Discharger) submitted a Report of Waste Discharge, dated 31 October 1996, and a Wastewater Treatment Facilities Phase 1 Improvements report, dated 16 December 1996, in support of a proposed discharge of 1.40 million gallons per day (mgd) of treated municipal wastes from its Wastewater Treatment Facility (WWTF) to irrigated land, proposed modifications to its WWTF, and the discharge of up to 1.8 mgd of subsurface drainage water from the irrigated land to a surface drainage channel (drainage channel). The Discharger provides sewerage service to City residents and industry. The WWTF is owned and operated by the Discharger, on property owned by the Discharger.
2. The Discharger currently discharges to an agricultural drainage ditch which discharges to Los Banos Creek, a water of the United States and a tributary to Mud Slough (north) and the San Joaquin River (above the Merced River). The discharge into Los Banos Creek is currently governed by Waste Discharge Requirements (WDRs) Order No. 96-193 (National Pollutant Elimination Discharge Permit No. CA0081272) and Cease and Desist Order No. 96-194, both adopted by the Board on 8 August 1996; and by an enforcement Order (Docket No. IXO FY-85-10), issued and amended by the EPA on 29 March 1985 and 10 December 1987, respectively.
3. The Discharger fails to meet the chronic and acute toxicity, pond dissolved oxygen, and receiving water limits of WDRs Order No. 96-193. It also fails to meet its total suspended solids (TSS) and total coliform organisms limits consistently. C&D Order No. 96-194 directs the Discharger to fully comply with WDRs Order No. 96-193, in particular with its toxicity and receiving water requirements, by eliminating the discharge to surface waters by 15 October 1998.
4. The existing WWTF includes a bar screen, two aerated ponds, nine oxidation ponds, six marsh cells, and chlorine disinfection and dechlorination units for a design flow of 1.2 million gallons per day (mgd). Present flows average about 1.0 mgd. The RWD indicates that about 55 percent of the flow and 70 percent of the organic loading is from Morningstar West and Beatrice Cheese Company, two milk processing facilities. Staff calculations indicate that from November 1996 through September of 1997, up to 90 to 97 percent of the organic loading may have been from industry.

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5. The Discharger completed a master plan in 1995 to upgrade and expand the capacity of its existing WWTF in phases from its current flow capacity of 1.2 mgd to 2.5 mgd to accommodate industrial and nonindustrial wastewater flows through the year 2014. As part of its strategy to comply with C&D Order No. 96-194, the Discharger proposes to eliminate its current year-round direct discharge to the agricultural drainage ditch and instead discharge seasonally to 220 acres of adjacent grassland for pasture of nonmilking animals and nesting bird habitat. The proposed reuse area is adjacent to and southwest of the WWTF, between Carnation Road and Gun Club Road. These proposed requirements address the proposed discharge of 1.4 mgd of wastewater to land and the discharge of subsurface drainage water to the drainage channel.
6. Phase 1 of the proposal is expected to be completed in October 1998 and consists of a WWTF expansion to 1.4 mgd that includes:
 - Upgraded interpond piping and flow control structures for the existing ponds
 - Bypass piping to bypass marsh system and, if not needed, the chlorination facilities
 - New splashers-type surface aerators for Pond Nos. 1, 2, and 3
 - Relocation of existing aerators from Pond Nos. 1 and 2 to Pond Nos. 4, 5, and 6
 - Two new effluent storage ponds, each with a storage capacity of 330-acre feet
 - A new pumping station and conveyance piping to transfer effluent from the treatment facilities to the new storage ponds
 - A new 220-acre reuse area comprised of Field Nos. 1, 2, and 3
 - An irrigation pumping station and associated piping to discharge stored effluent to the new reuse area
 - A tailwater collection and return system for the reuse area
 - A subsurface drainage system for the reuse area consisting of a subsurface drainage sump, a pump station, and a network of drains.
7. The subsurface drainage sump and pump station will be installed at the northeast corner of Field No. 3. The proposed drains range from 6 to 12 inches in diameter and will be installed at a depth of approximately 8 feet below Field Nos. 2 and 3. A schematic of the modified WWTF is presented in Attachment C.
8. The Discharger proposes to use dewatered pond sludge on City-owned farmland as a soil amendment.
9. The Discharger is pursuing a Clean Water Replacement Grant for its proposed upgrades, and the Board recommended the Discharger to the State Water Resources Control Board for a low interest loan to assist in financing of the expansion. The Discharger is currently on the SWRCB list to receive a loan.

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10. The WWTF, proposed storage ponds, and 220-acre reuse area are in Section 10, T8S, R9E, MDB&M, with surface water drainage to Los Banos Creek, as shown in Attachments A and B, which are attached hereto and part of this Order by reference. The drainage channel discharges to the Westside Ditch, which discharges to Los Banos Creek approximately 1.1 channel miles east of the WWTF. Los Banos Creek discharges to Mud Slough approximately 4.1 channel miles downstream of the Los Banos Creek/drainage channel confluence. Mud Slough, in turn, discharges to the San Joaquin River approximately 3.5 miles downstream of the Los Banos Creek/Mud Slough confluence. Vernalis, on the San Joaquin River, is approximately 49 miles downstream of the Mud Slough/San Joaquin River confluence. It would take approximately three days for discharged drainage water to travel from the Los Banos Creek/Mud Slough confluence to Vernalis. The site lies within the Grassland Watershed and the Los Banos hydrologic area (No. 541.20), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
11. Borings conducted in October 1996 at the proposed storage pond site indicate that its soils are uniform and consist of fine sandy, silty clays, and clayey silts to a depth of 21.5 feet bgs. The permeability of these soils was determined to be 2×10^{-7} to 4×10^{-7} cm/sec.
12. The Board adopted a *Water Quality Control Plan, Third Edition, for the Sacramento River Basin and San Joaquin River Basin*, (hereafter Basin Plan) which designates beneficial uses, establishes water quality objectives (WQOs), and contains implementation plans and policies for waters of the Basin. These requirements implement the Basin Plan.
13. Los Banos Creek and Mud Slough carry agricultural drainage, excess irrigation water, and seasonal precipitation to the San Joaquin River. The beneficial uses of Los Banos Creek are agricultural supply, warm freshwater habitat, and wildlife habitat. The beneficial uses of Mud Slough are agricultural supply; water contact and noncontact water recreation; warm freshwater habitat; spawning, reproduction, and/or early development; wildlife habitat; commercial and sport fishing; and shellfish harvesting. The beneficial uses of the San Joaquin River downstream of the discharge are potential municipal and domestic supply, industrial process supply, agricultural supply, water contact and noncontact water recreation, warm freshwater habitat, migration of aquatic organisms; spawning, reproduction, and/or early development; and wildlife habitat.
14. The beneficial uses of the underlying groundwater are municipal and domestic, industrial, and agricultural supply.
15. Local groundwater occurs in confined and unconfined aquifers. The unconfined aquifer begins at 250 feet below ground surface (bgs), but there are shallow and perched groundwater bodies that may not be connected to the unconfined aquifer. Shallow groundwater beneath the project area occurs 2 to 6 feet bgs and is of moderate quality with concentrations of total dissolved solids (TDS) of about 421 to 2515 mg/l, a conductivity at 25°C (EC) of 660 to 2510 μ mhos/cm, boron concentrations of 0.18 to 9.8 mg/l, and selenium concentrations from 0.7 to 3.6 μ g/l. Shallow

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groundwater is not currently used for municipal or domestic purposes. However, shallow groundwater is pumped into irrigation channels for groundwater level control, and in the channels it blends with irrigation water and water that supports wildlife habitat. Excess irrigation water eventually reaches Los Banos Creek. Source water for the City is provided by wells and is of good quality with concentrations of total dissolved solids (TDS) of about 680 to 840 mg/l, an EC of 1100 to 1300 μ mhos/cm, and nitrate concentrations of 29 to 41 mg/l. The average selenium concentration in the City's supply wells is about 0.8 μ g/l.

16. As depicted in Attachment C, a flow schematic, wastewater from the treatment system will be discharged year-round to storage ponds. During the irrigation season (March through October), the Discharger proposes to pump effluent (chlorinated as required) that has accumulated during the non-irrigation season from the two storage ponds to irrigate pasture and grassland habitat in the reuse area by the graded-border surface irrigation method. All irrigation tailwater will be recovered and returned to the storage ponds for reapplication. From November through February, wastewater application will normally cease and storm water runoff from the reuse area will be discharged to the drainage channel, provided it has been at least three days since the last application of reused water. If there has been less than three days between the last application of reused water, storm water runoff will be captured in the tailwater return system and returned to the storage ponds. The Discharger also proposes to install a subsurface drainage system to control salinity levels in the root zone of the reuse area, maintain crop yield, and protect shallow groundwater quality. From November through March 15, it proposes, under a worst-case scenario, to discharge up to 1.8 mgd, for a total of 420 acre-feet annually, of subsurface drainage water generated by the reuse area into the drainage channel adjacent to and east of the proposed reuse area. Under normal conditions, the Discharger expects to discharge from the subsurface drainage system about every three years to manage the root zone saturation and salinity. As previously mentioned, the drainage channel into which subsurface drainage will be discharged discharges into Los Banos Creek approximately one mile east of the WWTF. Los Banos Creek is typically not used for irrigation purposes during the proposed period of discharge of subsurface drainage water. The projected quality of WWTF effluent, effluent discharged to the reuse area, and subsurface drainage water is as follows:

<u>Constituent</u>	<u>WWTF Effluent</u>	<u>Applied Effluent</u>	<u>Subsurface Drainage</u>
BOD (mg/l)	45	40	< 1
TSS (mg/l)	90	60	< 2
Ammonia-N (mg/l)	< 15	< 5	< 0.5
Nitrate-N (mg/l)	< 1	< 1	< 1

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<u>Constituent</u>	<u>WWTF Effluent</u>	<u>Applied Effluent</u>	<u>Subsurface Drainage</u>
TDS (mg/l)	1670	2309	5494
EC (µmhos/cm)	2778	3842	9140
Boron (mg/l)	0.9	1.24	2.96
Selenium (µg/l)	0.83	0.83	1.97

17. More recent data from the Discharger's self monitoring reports indicates that from May 1997 through September 1997 the monthly average EC of the effluent ranged from 5606 µmhos/cm to 7812 µmhos/cm. Daily values as high as 10000 µmhos/cm were recorded during May, July, and August of 1997. An effluent grab sample collected from the pond system on 22 November 1996 had sodium and chloride concentrations of 570 mg/l and 640 mg/l, respectively. The more recent data confirm that the wastewater influent is industry dominated and that effluent contains abnormal quantities of NaCl salt.
18. Wastewater discharge to the reuse area will overlap the nesting period for waterfowl and waterbirds, which is February through September. Waterbirds and waterfowl using the reuse area for nesting bird habitat could experience impacts from avian botulism and fowl cholera. A waterfowl and waterbird surveillance and control program is necessary to monitor for and mitigate potential impacts.
19. The Board amended the Basin Plan on 3 May 1996 to control agricultural subsurface drainage discharges. The amendment includes designation of beneficial uses, WQOs, and an implementation plan for achieving the WQOs for the Grassland Watershed water bodies. The Grassland Watershed water bodies include Mud Slough (north), Salt Slough, and water channels for the distribution of water supplies to the wetlands of the Grassland Watershed. The amendment also provides for the Board to adopt effluent limits in waste discharge requirements for discharges of subsurface drainage water from the Grassland Watershed to meet WQOs. For Mud Slough, the maximum and monthly average (15 March through 15 September) WQOs for Boron are 5.8 and 2.0 mg/l, respectively; the maximum and monthly mean WQOs for molybdenum are 0.050 and 0.019 mg/l, respectively; and the 4-day average WQO for selenium, which becomes effective on 1 October 2010, is 5 µg/l. For Los Banos Creek, the daily maximum and monthly mean WQO concentrations for selenium are 20 µg/l and 2.0 µg/l, respectively.
20. The Basin Plan and its amendments prohibit certain activities associated with subsurface drainage water discharges. Specifically, Prohibition 6.a, states, in part:

"The discharge of agricultural subsurface drainage form the Grassland watershed to the San Joaquin River or its tributaries from ... any subsurface drain ... is prohibited, unless such discharge is governed by waste discharge requirements."

Prohibition 6.e, states:

"Activities that increase the discharge of poor quality agricultural subsurface drainage are prohibited."

The proposed discharge complies with Prohibition 6.a, as the discharge will be regulated by this Order. Prohibition 6.e does not apply to the Discharger's proposed subsurface drainage water discharge because the discharge will not result in an increase in the discharge of "poor quality" subsurface drainage. Prohibition 6.e originated as a policy (Policy IV) statement in the amendments to the Basin Plan that were approved by the Board on 8 December 1988. In the 1994 edition, this statement appears in both the policy section and the prohibition section. "Poor quality" is not defined in the Basin Plan, nor was it defined in the 1988, 1994, 1995, or 1996 amendments. The staff report supporting the 1988 Basin Plan amendments is the only document that describes "poor quality" for purposes of interpreting this particular policy. Pursuant to this description, activities that increase the discharge of "poor quality" agricultural subsurface drainage include the construction of new subsurface drainage facilities, or the expansion of existing facilities, in areas where the drainage already has an average selenium concentration of 20 µg/l, and where drainage already has an average selenium concentration greater than 5 µg/l unless certain conditions are met. It is, therefore, obvious that "poor quality" refers to discharges of subsurface drainage water containing selenium concentrations in excess of 5 µg/l. There is no indication that the Board, through subsequent amendments to the Basin Plan, intended to modify the meaning of "poor quality" in any way. The proposed discharge will be low in selenium with expected selenium concentrations below 5 µg/l.

21. The Basin Plan states that groundwaters designated for use as municipal or domestic supply shall not contain concentrations of chemical constituents in excess of the Maximum Contaminant Levels (MCLs) specified in Title 22, California Code of Regulations (CCR), Division 4, Chapter 15 (hereafter Title 22). The recommended MCL for EC is 900 µmhos/cm.
22. The Basin Plan also requires that waters designated for use as agricultural supply shall not contain chemical constituents in amounts that adversely affect that use. The United States Department of Food and Agriculture has published guidelines for irrigation waters which indicate that irrigation waters with EC values and chloride concentrations of 1000 µmhos/cm and 170 mg/l, respectively, are good to excellent and suitable for irrigation of most plants under most conditions.
23. The "Sources of Drinking Water" policy, which is part of the Basin Plan, provides that all groundwater is considered to be suitable, or potentially suitable, for municipal or domestic water supply, and should remain so designated with certain exceptions. One of those exceptions is for water where the TDS exceeds 3000 mg/l (EC of 5000 µmhos/cm) and it is not reasonably expected to supply a public water system. Based on Finding No. 15, above, groundwater beneath the proposed reuse area is considered to be suitable or potentially suitable for municipal and domestic supply.

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24. The Gustine Drainage District (GDD) is the agency responsible for collection and disposal of areal surface and subsurface drainage. GDD also discharges its drainage into tributaries to Los Banos Creek. The discharge of drainage wastes from GDD's facilities into surface waters is not currently regulated under waste discharge requirements.
25. The Discharger submitted an Antidegradation Analysis dated 6 March 1996 which indicates that typical EC levels in Los Banos Creek, above the proposed discharge point for the subsurface drainage water, range from 710 to 2400 $\mu\text{mhos/cm}$. The report also indicates that EC levels in Mud Slough above its confluence with Los Banos Creek have been recorded as low as 510 $\mu\text{mhos/cm}$ and as high as 15900 $\mu\text{mhos/cm}$. According to the report, low flows in Los Banos Creek above the WWTF have been estimated at 10 cubic feet per second (cfs) from January through March.
26. Under the Discharger's current proposal, if effluent from the storage ponds is applied to the reuse area at volumes exceeding evapotranspiration rates, it could leach salts at concentrations that threaten WQOs (e.g., MCLs) in shallow groundwater. According to the 6 March 1996 report, the proposed subsurface drains will essentially prevent migration of pollutants from the reuse area at concentrations that threaten the quality of groundwater below the drains. Groundwater monitoring wells are necessary to assess the overall effectiveness of the subsurface drain system, determine compliance with groundwater WQOs, and ensure that any impact that disposal operations have on all groundwater resources is insignificant.
27. The Discharger submitted a report entitled, "*Receiving Water Study, July-August 1992*," that indicates that the average flow rate and EC level in Los Banos Creek upstream of its confluence with the unnamed drainage channel were 0.48 cfs and 2163 $\mu\text{mhos/cm}$ respectively for July and August 1993. The Discharger also submitted a report entitled "*Receiving Water Study, February-March 1993*" that the average flow and EC concentration in Los Banos Creek upstream of its confluence with the unnamed channel were 138 cfs and 1430 $\mu\text{mhos/cm}$, respectively, during February and March 1993. The anti-degradation analysis provided by the Discharger in its 6 March 1996 report indicates that the proposed discharge of subsurface drainage water into Los Banos Creek by itself could increase the EC of the creek to 3800 $\mu\text{mhos/cm}$ when the flow rate and EC level in the creek are 10 cfs and the 3400 $\mu\text{mhos/cm}$, respectively. Staff analysis of the flow and EC data for the creek indicate that significant degradation, particularly under drought conditions, could also occur under the flow and EC conditions identified in the receiving water studies. Accordingly, both analyses show that there is reasonable potential for exceeding background EC and adversely affecting the beneficial uses and water quality in Los Banos Creek.
28. The Discharger proposes to mitigate this potential degradation of water quality in Los Banos Creek by (a) requiring its industrial users to develop and implement salinity control plans to reduce effluent EC concentrations, which in turn should significantly reduce the EC of the subsurface

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drainage water; and/or (b) discharging the subsurface drainage water into Los Banos Creek when flows in Los Banos Creek are at least 14.5 cfs. However, even when flows are 14.5 cfs, without reduced ECs the discharge could increase the EC levels in the creek from 710 to about 2050 $\mu\text{mhos/cm}$.

29. Subsequent to the studies and analyses conducted by the Discharger, the U.S. Bureau of Reclamation (Bureau) and drainage dischargers in the Grasslands Subarea (Bureau designation) implemented the Grassland Bypass Project. The bypass utilizes a 28-mile segment of the San Luis Drain (Grassland Bypass) to convey agricultural drainage waters generated in the Grassland Subarea to the San Joaquin River via a six-mile segment of Mud Slough (north). As a result, drainage waters that were previously conveyed to the San Joaquin River via Mud and Salt Sloughs are now concentrated in Mud Slough (north).
30. Mud Slough (north) is listed in the Basin Plan as a Water Quality Limited Segment for selenium, TDS, pesticides, and toxicity. The Basin Plan also lists the San Joaquin River from Sack Dam to Vernalis as a Water Quality Limited Segment for selenium, toxicity, diazinon, parathion, carbaryl, eptam, Group A, toxaphene, DDT and endosulfan. Group A refers to the following chemicals, either individually or in combination: aldrin, dieldrin, endrin, heptachlor, heptachlor epoxide, chlordane (total), hexachlorocyclohexane (total), endosulfan (total) and toxaphene. The Basin Plan also designates numerical WQOs for salinity in the San Joaquin River at Vernalis. The 30-day running average EC objective from April to August is 700 $\mu\text{mhos/cm}$. The 30-day running average EC objective from September to March is 1000 $\mu\text{mhos/cm}$. From October of 1985 to September of 1996, the April to August EC objective was violated 58% of the time and the September to March salinity objective was violated 13% of the time. Salt loading resulting from the City of Gustine's proposed discharges of subsurface drainage water have the potential to contribute to the violations of the WQOs in Mud Slough and the San Joaquin River at Vernalis.
31. Based on the analyses and information presented in Finding Nos. 21 through 30, above, additional salinity controls on the discharge, such as effluent EC reductions and timing restrictions for the proposed discharges and/or application of low EC reclaimed water at evapotranspiration rates, are necessary to prevent adverse impacts on water quality in Los Banos Creek, Mud Slough, and the San Joaquin River; to help meet salinity WQO in the San Joaquin River at Vernalis; and to comply with the antidegradation provisions of the State Water Resources Control Board Resolution No. 68-18. Monitoring of the discharge of subsurface drainage water is necessary to verify compliance with WQOs, Resolution No. 68-18, and this Order.
32. Federal regulations for storm water discharges were promulgated by the EPA on 16 November 1990 (40 CFR Parts 122, 123, and 124). In response to the regulations, the State Water Resources Control Board adopted Water Quality Order No. 97-03-DWQ, National Pollutant Discharge Elimination System (NPDES) General Permit No. CAS000001 (General Permit), Waste Discharge

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Requirements for Dischargers of Storm Water Associated with Industrial Activities Excluding Construction Activities. The Discharger's WWTF (SIC Code No. 4952) will have a treatment capacity in excess of 1.0 mgd. Therefore, this Order requires the Discharger to obtain coverage under the General Permit for the discharges of storm water associated with industrial activities from its WWTF. Alternatively, the City can certify that all storm water is contained on-site. The Discharger maintains a separate storm drainage collection system for the City that conveys storm water to duck clubs about 0.75 miles east of the WWTF. The storm drain system is also used to collect discharges of municipal swimming pool wastewater and industrial cooling wastewater, which are not believed to contain significant pollutants. The discharge of storm water from the City storm drain system is not currently subject to regulation and will not be regulated by this Order or the General Permit.

33. The discharge of subsurface agricultural drainage into surface waters is exempt from the National Pollutant Discharge Elimination System permitting requirements of the Clean Water Act.
34. The Discharger is required to protect the environment to the greatest degree possible (Public Resources Code, Section 21000, et seq.). Based on the significant industrial component of the total flow, particularly salt concentrations that cause effluent EC to be well above typical municipal wastewater; and the seasonal contribution of salt to a Water Quality Limited Segment, a pretreatment program is appropriate pursuant to 23 CCR 2235. The purpose of a required program would be to assure that the significant industries implement best practicable treatment to protect the groundwater and surface water from pollutants, such as salt, that the WWTF cannot treat.
35. Order No. 96-193 finds that the Discharger adopted Ordinance No. 377 to implement an industrial pretreatment program to control industrial user discharges into its collection system and WWTF. Order No. 96-193 required the Discharger to submit by 15 October 1996 for approval (a) its ordinance to assure that it contained the necessary legal authorities to enforce the general pretreatment standards of Title 40, Code of Federal Regulations (CFR), Part 403.5; and (b) technical justification for local limits developed for conventional pollutants. The Discharger only recently submitted the Ordinance but never the justification for its local limits. The Discharger does not currently impose EC limits on its industrial dischargers.
36. The discharge is municipal wastewater with a significant industrial component, and it contains total dissolved solids (TDS), bacteria, nitrates, metals, organics, and other waste constituents. The treatment process to be employed by the Discharger will be the best practicable treatment and control for this wastewater, provided it employs effective salinity source controls.

Surface Water: The triennial subsurface drainage water discharge will replace the existing continuous NPDES discharge to Los Banos Creek. As such, the salts discharged to Los Banos Creek from the drainage system, up to presently permitted flows (1.2 mgd), will not represent a

new salt load on the San Joaquin River system, but will instead replace the existing salt load associated with the NPDES discharge. The implementation of salinity controls on industrial users and the new discharge scheme will reduce the overall annual mass salt load and limit it to a time of the year when receiving waters have the most assimilative capacity. Nevertheless, salts in the subject discharge do have the potential to degrade the waters in Los Banos Creek, Mud Slough, and contribute to violations of WQOs in the San Joaquin River. Under the terms of this Order, the Discharger must implement effective source control measures to effect best practicable technology before the Board can determine that it does not cause or unreasonably contribute to the exceedence of WQOs of these waters. With these provisions, and the fact that the proposed discharge allows wastewater utility service necessary to accommodate housing and economic expansion in the area, a benefit to the people of the State, the antidegradation provisions of State Water Resources Control Board Resolution 68-16 are met.

Groundwater: The treatment of wastewater in ponds and discharge of effluent to storage ponds and land, with some disposal by percolation, could result in increases in concentrations of some of these constituents in groundwater in the general proximity of the discharge. Some increase in pollutant concentrations locally in groundwater to allow wastewater utility service that is necessary to accommodate housing and economic expansion in the area is in the best interests of the people of the state of California. The impermeability of the underlying soil and subsurface drainage system in the reuse area will minimize the amount and dispersion of percolating effluent. A localized increase in the concentration of constituents in groundwater is consistent with the antidegradation provisions of State Water Resources Control Board Resolution No. 68-16, provided groundwater quality limits are met at on-property "points of compliance" (POCs) determined by the Board and the Discharger is employing best practicable treatment technology. To remain consistent with these provisions, increases in constituent concentrations cannot exceed groundwater quality limits at POCs. The POCs are established herein as part of the required monitoring well network to effectively monitor compliance with groundwater quality limits.

37. The City of Gustine approved a negative declaration for proposed project on 28 June 1996 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The Board has reviewed the negative declaration and compliance with these waste discharge requirements will mitigate or avoid potential adverse water quality and resources impacts.
38. The California Department of Health Services (DHS) has established statewide reuse criteria in Title 22, California Code of Regulations (CCR), Section 60301, et seq., (hereafter Title 22). Pursuant to Title 22, CCR, Section 60323, the Discharger's consultant submitted a water reuse engineering report dated 27 August 1997 (hereafter Title 22 Report) to DHS. The DHS approved the report by letter dated 8 September 1997. The Board has consulted with DHS, and incorporated DHS's recommendations into this Order.

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39. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
40. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the City of Gustine, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, and the provisions of the Clean Water Act and regulations and guidelines adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Effective **15 October 1998**, discharge of wastewater effluent directly from the WWTF to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. The discharges of wastewater and tail water from the reuse area to surface waters or surface water drainage courses is prohibited.
4. Discharge of subsurface drainage water from the reuse area subsurface drainage system is prohibited from **15 March through October**.
5. Discharge of waste classified as 'hazardous,' as defined in Section 2521(a) of Title 23, CCR, Section 2510, et seq., or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications

1. The monthly average dry weather (May through October) discharge to the storage ponds after the treatment ponds shall not exceed 1.40 mgd (See Provision No. H.1).
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and reuse areas to a degree that threatens nuisance.
3. As a means of discerning compliance with Discharge Specification No. B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.

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4. Effluent from the WWTF shall not exceed the following limits:

<u>Constituents</u>	<u>Units</u>	<u>Monthly Average</u>	<u>7-Day Median</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	40	--	80
Settleable Solids	ml/l	0.2	--	0.5

¹ 5-day, 20°C biochemical oxygen demand (BOD)

5. The discharge from the treatment ponds shall not have a pH less than 6.5 or greater than 9.0 pH units.
6. The Discharger shall make every reasonable effort to lower its influent EC to a level that primarily reflects the EC associated with the typical municipal wastewater. Until the Discharger documents the actual sources of EC increases in its wastewater, the interim influent EC performance goal shall be 1,800 µmhos/cm (See Provision No. H. 14). Reasonable effort shall include, but not be limited to, requiring industry to employ best practicable treatment to achieve the EC performance goal. Exceedences of the performance goal shall not be considered a violation of this Order, but shall be reported to the Board, with monthly self monitoring reports along with an explanation for the exceedences.
7. Ponds shall be managed to prevent breeding of mosquitoes. In particular:
- An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - Dead algae, vegetation, and debris shall not accumulate on the water surface.
8. Public contact with wastewater shall be precluded through such means as fences and signs, or acceptable alternatives.
9. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season, excluding wastewater discharged as authorized by valid requirements to surface waters, reuse sites, and off-site wastewater storage and disposal areas. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed hly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically) or a lesser freeboard if certified in writing by a registered civil engineer as adequate to prevent overtopping, overflows, or levee failures.

10. On or about 15 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.9.

C. Subsurface Drainage Water Specifications

1. The maximum daily discharge flow rate of subsurface drainage water shall not exceed 1.8 mgd.
2. The discharge of subsurface drainage water to Los Banos Creek shall be limited to the period from 1 November through 15 March and shall be timed to occur when the salinity objective for the San Joaquin River at Vernalis is being met and the River has sufficient capacity to assimilate any additional salt loading resulting from the discharge without violating WQOs, and when Los Banos Creek and Mud Slough have sufficient capacity to assimilate any additional selenium and boron loading resulting from the discharge without violating WQOs.

D. Reuse Specifications

1. Reused water shall remain within the 220-acre designated reuse area except as returned to the storage reservoirs as irrigation return water.
2. Discharge of rainwater runoff from the reuse area shall only occur when at least one day has passed since the last application of reused water.
3. The Discharger's use of reused water in the 220-acre reuse area shall be limited to irrigation of pasture for nonmilking beef animals, wildlife habitat, and/or fodder, fiber, and seed crops, and shall comply with the provisions of Title 22 and the approved Title 22 report.
4. The total volume of reused water applied shall not exceed the amount shown to be reasonable in an approved irrigation management plan, using accepted crop management practices.
5. Water reuse shall comply with the Title 22 report approved by the DHS.
6. Areas irrigated with reused water shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within a 48-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.

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7. Reused water shall be managed to prevent runoff onto adjacent properties not owned or controlled by the Discharger.
8. All areas where reused water is to be used shall be posted with conspicuous signs with the following wording in a size that can be clearly read by the public: "RECLAIMED WATER--DO NOT DRINK--WASH THOROUGHLY WITH SOAP AND DRINKING WATER IF CONTACT OCCURS." Where appropriate, bilingual signs shall be used.
9. No physical connection shall exist between reused water piping and any domestic water supply or domestic well, or between reused water piping and any irrigation well that does not have an air gap or reduced pressure principle device.
10. Reused water shall be managed to minimize contact with workers.
11. If reused water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes." Other uses of reused water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with Title 22.
12. Bird carcasses shall be burned or buried unless an unusual number (more than 15) is found. Upon finding an unusual amount, the California Department of Fish and Game (DFG) shall be notified within 24 hours and bird carcasses shall, at the DFG's discretion, be held for diagnosis. A record of the number, date, and species of carcasses found and burned or buried shall be kept.

E. Sludge Disposal Specifications

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, California Code of Regulations, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer **at least 60 days in advance** of the change.
3. Use and disposal of sewage sludge shall comply with existing federal, state, and local laws, regulations, and ordinances, including permitting requirements and technical standards included in 40 CFR 503 and the requirements of Merced County Sludge Ordinance #1505.

If the State Water Resources Control Board and the Regional Water Quality Control Boards assume primacy to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

F. Receiving Water Limitations

Receiving water limitations are based upon water quality objectives contained in the Basin Plan. As such, they are a required part of this Order. The discharge of subsurface drainage water from the WWTF to the drainage channel shall not cause or contribute to the following in Los Banos Creek:

1. Concentrations of dissolved oxygen to fall below 7.0 mg/l.
2. Oils, greases, waxes, or other materials to form a visible film or coating on the water surface or on objects in the water.
3. Oils, greases, waxes, floating material (liquids, solids, foams, and scums) or suspended material in amounts that create a nuisance or adversely affect beneficial uses.
4. Chlorine to be in detectable concentrations.
5. Pesticide or combination of pesticides to be detected in concentrations that adversely affect beneficial uses.
6. Aesthetically undesirable discoloration.
7. Fungi, slimes, or other objectionable growths.
8. Turbidity to increase more than one turbidity unit (NTU) when background levels are between 0 and 5 NTU; to increase 20% over background levels when the background levels are between 5 and 50 NTU; to increase 10 NTU when background levels are between 50 and 100 NTU; and to increase 10% when background levels are greater than 100 NTU.
9. The normal ambient pH to fall below 6.5, exceed 8.5, or change by more than 0.5 pH units.
10. Deposition of material that causes nuisance or adversely affects beneficial uses.
11. The normal ambient temperature to increase more than 5°F (2.8°C).

12. Taste or odor-producing substances to impart undesirable tastes or odors to the water or fish or other edible products of aquatic origin or to cause nuisance or otherwise adversely affect beneficial uses.
13. Radionuclides to be present in concentrations that exceed maximum contaminant levels specified in the California Code of Regulations, Title 22; that harm human, plant, animal or aquatic life; or that result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal or aquatic life.
14. The fecal coliform concentration in any 30-day period to exceed a geometric mean of 200 MPN/100 ml or cause more than 10 percent of total samples to exceed 400 MPN/100 ml.
15. Chemical constituents to be present in concentrations that adversely affect beneficial uses.
16. Toxic pollutants to be present in the water column, sediments, or biota in concentrations that adversely affect beneficial uses or that produce detrimental physiological responses in human, plant, animal, or aquatic life.
17. Violations of any currently applicable water quality standard for receiving waters adopted by the Board or the State Water Resources Control Board pursuant to the CWA and regulations adopted thereunder (See Provision H.20).

The daily maximum and monthly mean concentrations of selenium to exceed 20 µg/l and 2.0 µg/l, respectively, and/or cause or contribute to the following in Mud Slough (north):

1. By 2010, the maximum and 4-day average concentration of selenium to exceed 0.02 mg/l and 0.005 mg/l, respectively.
2. The maximum and monthly mean concentration of molybdenum to exceed 0.05 mg/l and 0.019 mg/l, respectively.
3. The monthly mean concentration of boron to exceed 2.0 mg/l from 15 March through 15 September and the maximum boron concentration to exceed 5.8 mg/l at any time.

G. Groundwater Limitations

The discharge shall not cause the underlying groundwater, as measured at POCs, to contain waste constituents in concentrations statistically greater than background water quality. For purposes of comparison, background water quality shall be determined when groundwater monitoring provides sufficient data (See Monitoring and Reporting Program No. 98-039 for implementation of this specification).

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H. Provisions

1. This Order is intended to supersede Waste Discharge Requirements (WDRs) Order No. 96-193 (National Pollutant Elimination Discharge Permit No. CA0081272. Use of the reuse area, specifications and limitations associated with the use of the reuse areas, and an increase in discharge flow above 1.2 mgd, become effective only upon the completion of construction of the proposed WWTF and written approval of the Executive Officer. To obtain written approval from the Executive Officer, the Discharger shall submit a written certification to the Board at least **30 days prior** to discharges to the completed WWTF improvements. The certification shall be appropriately signed and stamped by a California registered civil engineer and shall certify that all critical improvements have been completed as designed. (Following approval of the certification, Order No. 96-193 shall be scheduled for rescission by the Board.)
2. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that without treatment are essentially free of pollutants.
3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
4. The Discharger shall comply with Monitoring and Reporting Program No. 98-039 (MRP), which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
5. By **60 days prior to startup of the new facilities**, the Discharger shall submit a copy of the operation and maintenance manual (O&M manual) for the WWTF, including its proposed modifications. The O&M manual shall be prepared under the supervision of a California registered civil engineer and include descriptions of complete operational procedures for the WWTF that assure compliance with this Order. It shall also include a detailed description of how the Discharger will operate and monitor the subsurface drainage system and monitor Los Banos Creek, Mud Slough, and the San Joaquin River to ensure compliance with the terms of this Order. Further, the O&M manual must address the items in Standard Provision B.2.
6. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

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7. To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, the address and telephone number of the persons responsible for contact with the Board and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.
8. By **60 days prior to startup of the new facilities**, the Discharger shall either submit a Notice of Intent to comply with Water Quality Order No. 97-03-DWQ, National Pollutant Discharge Elimination System (NPDES) General Permit No. CAS000001 (General Permit), Waste Discharge Requirements for Dischargers of Storm Water Associated with Industrial Activities Excluding Construction Activities, or provide written certification that storm water associated with industrial activities at its WWTF will not under any circumstances leave the facility site.
9. The Discharger shall conduct the acute and chronic toxicity testing specified in Monitoring and Reporting Program No. 98-039. If toxicity is observed in any combination of acute and chronic toxicity tests resulting in two positive tests for toxicity during the discharge season, the Discharger shall submit, within 30 days of receiving the second test result, a work plan to conduct a toxicity reduction evaluation (TRE) and, upon approval, conduct the TRE. Alternatively, the Discharger may submit a work plan, subject to Executive Officer approval, to complete a biological study of the receiving water to determine if the discharge is causing adverse impacts to the local ecosystem. Based on the results of the TRE and/or the biological study, this Order may be reopened to include toxicity limitations and/or limitations for specific toxicants. Additionally, if acute or chronic toxicity water quality objectives are adopted by the SWRCB, this Order may be reopened and a limitation based on the objectives included.
10. By **at least 120 days prior to startup of the new facilities**, the Discharger shall submit a technical report in the form of a work plan to install and develop a comprehensive ground water monitoring network to monitor at the earliest practicable date at the POCs the effects of its disposal operations on the uppermost groundwater aquifer. The proposed program shall include, but not be limited to: (1) a sufficient number of upgradient and downgradient ground water monitoring wells for the reuse area and storage ponds, (2) a map showing the proposed location of the wells, and (3) the proposed well specifications. The Discharger shall install the network **within 90 days** of the Executive Officer's approval of the work plan and monitor all monitoring wells as described in the attached Monitoring and Reporting Program.

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11. By at least 60 days prior to startup of the new facilities, the Discharger shall submit an engineering report in the form of final irrigation management plan for the reuse area that addresses compliance with Reuse Specification Nos. D.1 through D.12.
12. The Discharger shall implement the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accommodate such wastes;
 - c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc.), released in such volume or strength as to cause inhibition or disruption in the treatment works, and subsequent treatment process upset and loss of treatment efficiency;
 - e. Heat in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40 °C (104°F), unless the treatment works is designed to accommodate such heat;
 - f. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;
 - g. Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and
 - h. Any trucked or hauled pollutants, except at points predesignated by the Discharger.
13. The Discharger shall implement the legal authorities, programs, and controls necessary to ensure that indirect discharges do not introduce pollutants into the sewerage system that, either alone or in conjunction with a discharge or discharges from other sources:
 - a. Flow through the system to the receiving water in quantities or concentrations that cause a violation of this Order, or

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- b. Inhibit or disrupt treatment processes, treatment system operations, or sludge processes, use, or disposal and either cause a violation of this Order or prevent sludge use or disposal in accordance with this Order.
14. In addition to Provision Nos. 12 and 13 and to comply with Discharge Specification No. B.6, the Discharger shall develop and implement a Pretreatment Program, subject to the approval of the Executive Officer. The Pretreatment Program shall become an enforceable condition of this Order. To satisfy this Provision, the Discharger shall comply with the following time schedule:

<u>Tasks</u>	<u>Compliance Date</u>
a. Submit the results of an industrial user survey.	1 May 1998
b. Submit proposed local limits for at least EC and total dissolved solids and a proposed Salinity Source Control Program Report (SSCPR). The SSCPR shall evaluate the contribution from industry and residential self-regenerating water softeners of salts to the influent waste stream and shall provide technical justification for proposed local limits. The SSCPR shall also include a summary of test results and average daily flows for each industrial contributor's wastewater, and a detailed description of the source control program proposed by the Discharger to regulate the discharge of salts to the treatment facility.	1 Sep 1998
c. Submit documentation for proposed Legal Authority and Control Mechanism implementation, an Enforcement Response Program, and a list of resource allotments for the program.	1 Oct 1998
d. Submit a proposed Inspection Monitoring Program and Pretreatment Ordinance, which will implement the requirements of the pretreatment program.	1 Oct 1998
e. Submit a complete pretreatment program package with request for pretreatment program approval.	1 Nov 1998
f. Implement the approved Pretreatment Program.	60 Days after Approval

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15. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
16. The Board will review this Order periodically and will revise requirements when necessary.
17. The Discharger shall submit to the Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule.
18. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
19. The Discharger shall notify the Board **at least 120 days prior to:**
 - a. any significant introduction of pollutants into the WWTF by a new industrial user.
 - b. any substantial change in the volume or character of pollutants being introduced into the WWTF by a source introducing pollutants into the WWTF at the time of adoption of this Order, and
 - c. any planned physical alterations or additions to the WWTF, or changes planned in the Discharger's sludge use or disposal practice, where such alterations, additions, or changes may justify the application of waste discharge requirement conditions that are different from or absent in the existing waste discharge requirements, including notification of additional a disposal sites not reported during the waste discharge application process, or not reported pursuant to an approved land application plan.
20. Should the Board or the State Water Resources Control Board adopt or modify applicable effluent or receiving water quality standards, this Order may be reopened and revised to reflect the new standards. The Discharger shall comply with effluent standards and prohibitions within the time provided in the regulations that establish those standards or prohibitions, even if this Order has not yet been modified.
21. Based on the results yielded for Provision 14.b and proposals by the Discharger, this Order may be reopened and Discharge Specification No. B.6 revised to replace the performance goal with a numeric effluent limit.

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I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 27 February 1998.



GARY M. CARLTON, Executive Officer

WDH:wdh/fmc:2/27/98

REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 98-039

FOR
CITY OF GUSTINE
WASTEWATER TREATMENT FACILITY
AND
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INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent. Influent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	Meter	Continuous
BOD ₅ ¹	mg/l, lbs/day	24-hr. Composite ²	Weekly
Suspended Solids	mg/l, lbs/day	24-hr. Composite ²	Weekly
Conductivity @ 25°C ³	µmhos/cm	Grab	Daily

¹ Five day, 20°C BOD.

² Composite samples may consist of flow-proportioned grab samples.

³ Hereafter referred to as EC.

EFFLUENT MONITORING

Effluent samples shall be collected downstream of the chlorination facilities, if in use, or the last treatment pond, if chlorination facilities are not in use, just prior to discharge to the storage ponds. Effluent samples shall be representative of the volume and quality of the discharge. Samples collected from the outlet structure of ponds will be considered adequately composited. Time of collection of samples shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
pH	pH units	Grab	Weekly
Chlorine Residual	mg/l	Grab	Weekly
EC	µmhos/cm	Grab	Weekly

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<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Total Dissolved Solids	mg/l	Grab	Monthly
BOD ₅	mg/l, lbs/day	8-hr. Composite	Weekly
Standard Minerals ¹	mg/l	Grab	Quarterly
Heavy Metals ²	µg/l	8-hr. Composite	Yearly

¹ Standard minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

² Aluminum (EPA 200 series), Antimony (EPA 204.2), Arsenic (EPA 206.3), Beryllium (EPA 200.7 or 210.2), Cadmium (EPA 213.2), Total Chromium (EPA 218.4), Chromium VI (218.4 or 7196), Copper (EPA 220.2), Lead (EPA 239.2), Mercury (EPA 245.1 or 245.2), Nickel (EPA 200.7 or 249.2), Selenium (EPA 270.3), Silver (EPA 272.2), Thallium (EPA 200.7 or 279.2), and Zinc (EPA 200.7 or 289.2).

POND MONITORING

The freeboard shall be monitored for all ponds to the nearest tenth foot. Pond water monitoring shall include the following :

<u>Constituent</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency¹</u>
Freeboard	feet	Observation	Weekly
pH	pH units	Grab	Weekly
Dissolved Oxygen ²	mg/l	Grab	Weekly

¹ Should sampling indicate a violation or threatened violation of the terms of this Order or should the WWTF ponds produce objectionable odors, the monitoring frequency for the subject ponds shall be increased to daily until violations, threatened violations, and/or odor producing conditions are resolved.

² Samples shall be collected from each pond at a depth of one foot, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

Permanent markers (e.g., staff gages) shall be placed in the ponds with calibration indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall inspect the conditions of the ponds once per week and write visual observations of potential problems in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their locations; whether dead algae, vegetation, scum, or debris are accumulating on the

pond surface and their location; whether burrowing animals or insects are present. A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

THREE SPECIES CHRONIC TOXICITY MONITORING

Chronic toxicity monitoring shall be conducted to determine if the discharge of subsurface drainage water is contributing toxicity to Los Banos Creek. The testing shall be conducted as specified in EPA 600/4-91-002. Chronic toxicity samples shall be collected at the subsurface drainage sump prior to discharge to the drainage channel. Samples shall be 8-hour composites and shall be representative of the volume and quality of the discharge. Time of collection of samples shall be recorded. The tests must be conducted with concurrent reference toxicant tests. Monthly laboratory reference toxicant tests may be substituted upon approval. Both the reference toxicant and effluent tests must meet the test acceptability criteria specified in the EPA manual. If the test acceptability criteria are not achieved, the Discharger must re-sample and retest within 15 days. Chronic toxicity monitoring shall include the following:

Species: Pimephales promelas, Ceriodaphnia dubia, and Selenastrum capricornutum
Frequency: At least twice during the subsurface drainage water discharge season*.
Dilution Series:

	<u>Dilutions (%)</u>					<u>Controls</u>	
	<u>100</u>	<u>75</u>	<u>50</u>	<u>25</u>	<u>12.5</u>	<u>Drainage Water</u>	<u>Lab Water</u>
% Drainage Water	100	75	50	25	12.5	0	0
% Dilution Water ¹	0	25	50	75	87.5	100	0
% Lab Water ²	0	0	0	0	0	0	100

¹ Dilution water shall be receiving water from Los Banos Creek taken at a point not to exceed 100 feet upstream from the confluence of Los Banos Creek with the drainage channel (See Attachment A).

² Lab water shall meet EPA protocol requirements.

* Should the results show toxicity, the sampling frequency shall be increased to weekly until the toxicity is resolved.

SLUDGE MONITORING

Sludge disposal shall be at agronomic rates. Prior to on-site disposal, a composite sludge sample shall be analyzed for the following metals:

Cadmium	Lead
Chromium	Nickel
Copper	Zinc

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Analytical results shall be submitted to the Executive Officer. Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report. **By 31 January of each year (commencing in 1999), the Discharger shall submit a description of present and proposed disposal methods.** If more than one method is used, include the percentage of annual sludge production disposed by each method:

1. For **landfill disposal**, include a) the Regional Board's WDR number that regulates the landfill(s) used, b) the present classifications of the landfill(s) used, c) the names and locations of the facilities receiving sludge.
2. For **land application**, include a) the location of the site(s), b) the Regional Board's WDR numbers that regulate the site(s), c) a technical report analyzing application rates and procedures relative to Department of Health Services' Manual of Good Practices for Landspreading of Sewage Sludge and EPA's Process Design Manual for land application of Municipal Sludges and Title 23, California Code of Regulations, Section 2511 (f) and (d) subsequent uses of the land.
3. For **composting**, include a) the location of the site(s), and b) the Regional Board's WDR numbers that regulate the site(s).

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Standard Minerals ¹	mg/l	Annually ³
EC ²	µmhos/cm	Annually ³
Total Dissolved Solids	mg/l	Annually ³

¹ Standard minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

² If the source water is from more than one well, the EC shall be reported as a weighted average and include copies of supporting calculations.

³ Quarterly during the first 12 months following the adoption of this program, annually thereafter.

GROUNDWATER MONITORING

Pursuant to Provision No. H.10, the Discharger shall sample the approved groundwater monitoring wells monthly for one year, and analyze them for the parameters specified below. After one year of sampling the wells, data from these analyses shall be reported to the Board by the 20th day of the following month for use in establishing water quality protection standards. If subsequent sampling of background monitoring wells indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The downgradient wells near the facility property boundaries shall constitute "points of compliance (POCs)." In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) and depth to ground water (in feet and hundredths below ground surface) in all wells shall be measured on a quarterly basis and used to determine the gradient and direction of ground water flow. Water samples shall be collected from the wells and analyzed as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
EC	µmhos/cm	Grab	Quarterly
pH	pH units	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Standard Minerals ¹	mg/l	Grab	Quarterly

¹ Standard minerals as used in this program shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

Following each sampling event, the Discharger shall determine whether there is a statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger or the Board finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POCs, the Discharger shall notify the Board immediately, or acknowledge the Board's findings, and submit within 90 days, either a technical report with a plan and time schedule for implementing a verification monitoring program or a report demonstrating water quality protection standards were not exceeded. The verification monitoring program must be designed to verify that water quality protection standards have been exceeded and the horizontal and vertical extent of the impact.

ANNUAL LAND MANAGEMENT MONITORING

The Discharger shall submit a summary report on the land management operation after conclusion of each season. The report shall discuss net acreage irrigated and total water application over the season; the total wastewater applied in acre-feet; sludges and chemical fertilizers; and amount of nutrients

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removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading. The report for each year is due by **31 January** of the following year.

SUBSURFACE DRAINAGE MONITORING

A monitoring station shall be established at the subsurface drainage pump station to collect representative samples of the subsurface drainage water prior to discharge to the drainage channel. Subsurface drainage water monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u> ¹
Flow	mgd	Metered	Continuous
pH	pH units	Grab	Weekly
EC	µmhos/cm	Grab	Weekly
Total Dissolved Solids	mg/l	Grab	Monthly
Pesticides ³	mg/l	Grab	2/Season
Standard Minerals ⁴	mg/l	Grab	2/Season
Metals ⁵	µg/l	24-hr. Composite	2/Season
Acute Toxicity ⁶	% survival	24-hr. Composite	2/Season ⁷

¹ To be monitored at the specified frequency during discharges from the subsurface drainage system. First round samples for non-weekly sampling shall be collected within one week of the initiation of the discharge.

² Most probable number.

³ EPA methods 619, 8081, 8140, 8150 and 8318.

⁴ Standard minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

⁵ Aluminum (EPA 200 series), Antimony (EPA 204.2), Arsenic (EPA 206.3), Beryllium (EPA 200.7 or 210.2), Cadmium (EPA 213.2), Total Chromium (EPA 218.4), Chromium VI (218.4 or 7196), Copper (EPA 220.2), Lead (EPA 239.2), Mercury (EPA 245.1 or 245.2), Nickel (EPA 200.7 or 249.2), Selenium (EPA 270.3), Silver (EPA 272.2), Thallium (EPA 200.7 or 279.2), and Zinc (EPA 200.7 or 289.2).

⁶ Acute toxicity shall be determined using a 96-hour continuous flow acute toxicity test conducted in accordance with EPA/600/4-90/027F, or later amendment approved by Board staff. The bioassay shall be conducted on samples collected from the subsurface drainage system and prior to discharge to the drainage channel. Fathead minnow (*Pimephales promelas*) shall be used as the test species. Temperature and pH shall be recorded at the time of bioassay sample collection.

⁷ Should results show toxicity, sampling frequency shall be increased to weekly until the toxicity is resolved.

REUSE AREA MONITORING

The Discharger shall inspect the reuse areas at least weekly. In conducting the inspection, a log shall be kept of reuse area conditions. Log entries shall confirm the presence or absence of the following:

- a. breaches of berms
- b. standing wastewater,
- c. erosion conditions that threaten to create nuisance conditions or off-site discharges,
- d. off-site runoff of applied effluent.

Log entries shall include date, time and effectiveness of the measures employed to mitigate problem conditions observed during the weekly inspections. In addition, the Discharger shall survey the reuse area for dead, deformed or otherwise impaired water fowl and water birds, and record the species, quantity, and final disposition of any impaired water fowl and water birds observed during the inspection in the inspection log. Summaries of log entries shall be submitted with monthly monitoring reports.

RECEIVING WATER MONITORING

All receiving water samples shall be grab samples collected at a depth of 6 to 12 inches below the surface. Receiving water monitoring shall include at least the following:

<u>Station</u>	<u>Description</u>
R-1	100 feet upstream of the point of discharge into the drainage channel.
R-2	At least 100 feet downstream of the point of discharge to the drainage channel.
R-3	100 feet upstream of the confluence of the discharge drainage ditch and Los Banos Creek.
R-4	At least 100 feet downstream of the confluence of the discharge drainage ditch and Los Banos Creek at a point that ensures complete mixing.
R-V	Regional Board monitoring station at Vernalis

<u>Constituent</u>	<u>Units</u>	<u>Station</u>	<u>Sampling Frequency</u>
Flow	cfs & mgd	R1, R3, RV	Weekly ^{1,2}
Total Dissolved Solids	mg/l	R1, R2, R3, R4, RV	Monthly ¹
Conductance	µmhos/cm	R1, R2, R3, R4, RV	Weekly ^{1,2}
Selenium	µg/l	R1, R2, R3, R4	Monthly ¹

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<u>Constituent</u>	<u>Units</u>	<u>Station</u>	<u>Sampling Frequency</u>
Boron	mg/l	R1, R2, R3, R4	Monthly ¹
Dissolved Oxygen	mg/l	R1, R2, R3, R4	Monthly ¹
Turbidity	NTU	R1, R2, R3, R4	Monthly ¹
Temperature	°C	R1, R2, R3, R4	Monthly ¹
pH	pH units	R1, R2, R3, R4	Monthly ¹

¹ To be monitored during discharges of subsurface drainage.

² Should remote monitoring at Vernalis indicate that salinity objectives are being exceeded, the monitoring frequency shall be increased to daily until objectives are met.

REPORTING

Daily, monthly, and quarterly sampling results shall be submitted to the Board by the **20th of the month** following the period the samples are collected. Noncompliance with waste discharge requirements shall be reported as soon as discovered.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By 20 October of each year, the Discharger shall certify in writing as to whether available pond storage volume is at least that which is required to comply with Discharge Specification No. B.9.

By 31 January of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grades and general responsibilities of persons operating and maintaining the wastewater treatment plant (Standard Provision E.1).

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- b. The names and telephone numbers of persons to contact regarding plant emergency and routine situations.
- c. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
- d. A statement identifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- f. A summary of the reuse area operation discussing total water application over the season; the total wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading.
- g. The annual report shall include the information required by the Sludge Monitoring, Water Supply Monitoring, and Annual Land Management Monitoring sections of this Monitoring and Reporting Program.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement this monitoring program upon start-up of the new facilities, unless otherwise required in the Provisions.

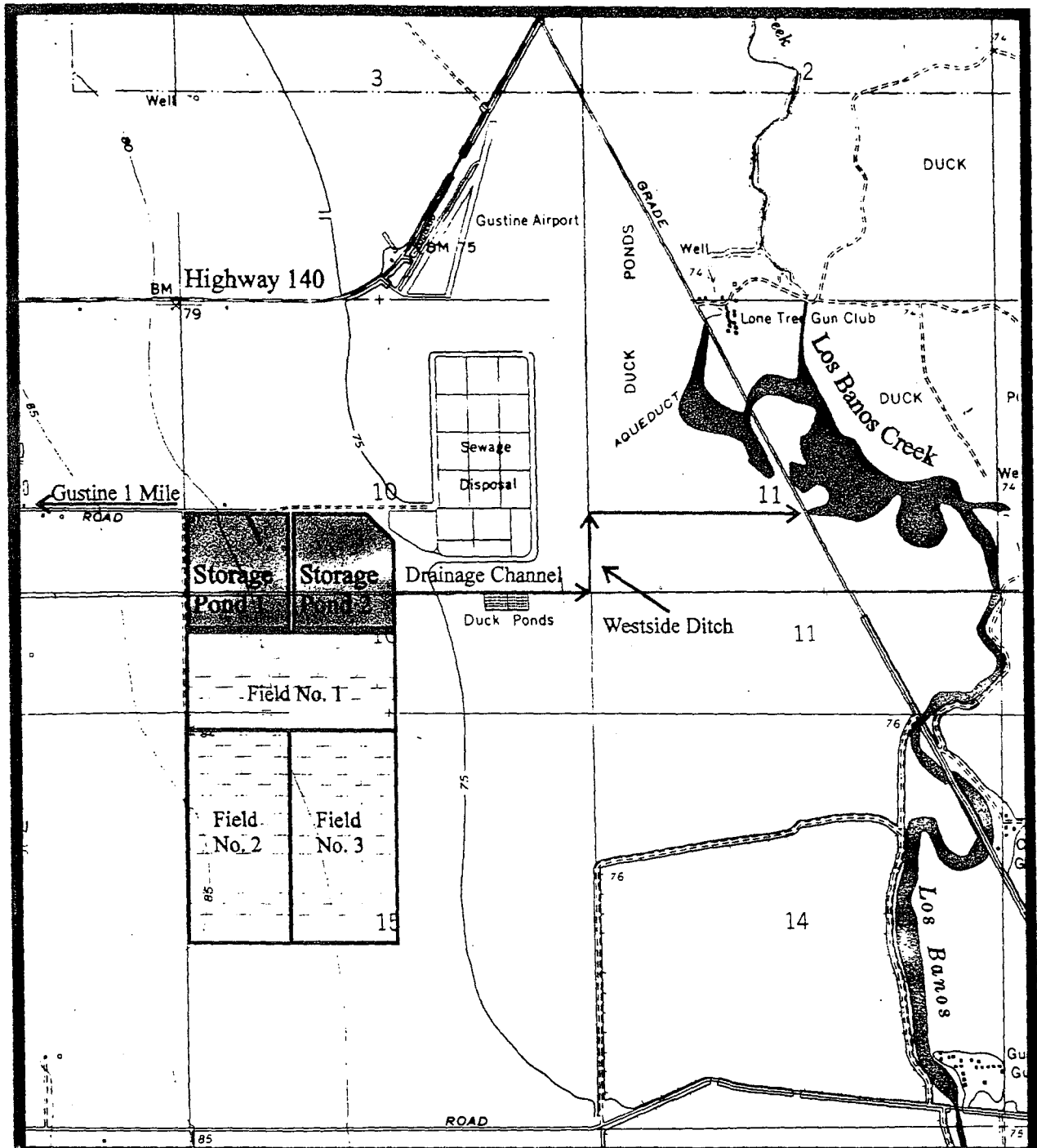
Ordered by:


GARY M. CARLTON, Executive Officer

27 February 1998

(Date)

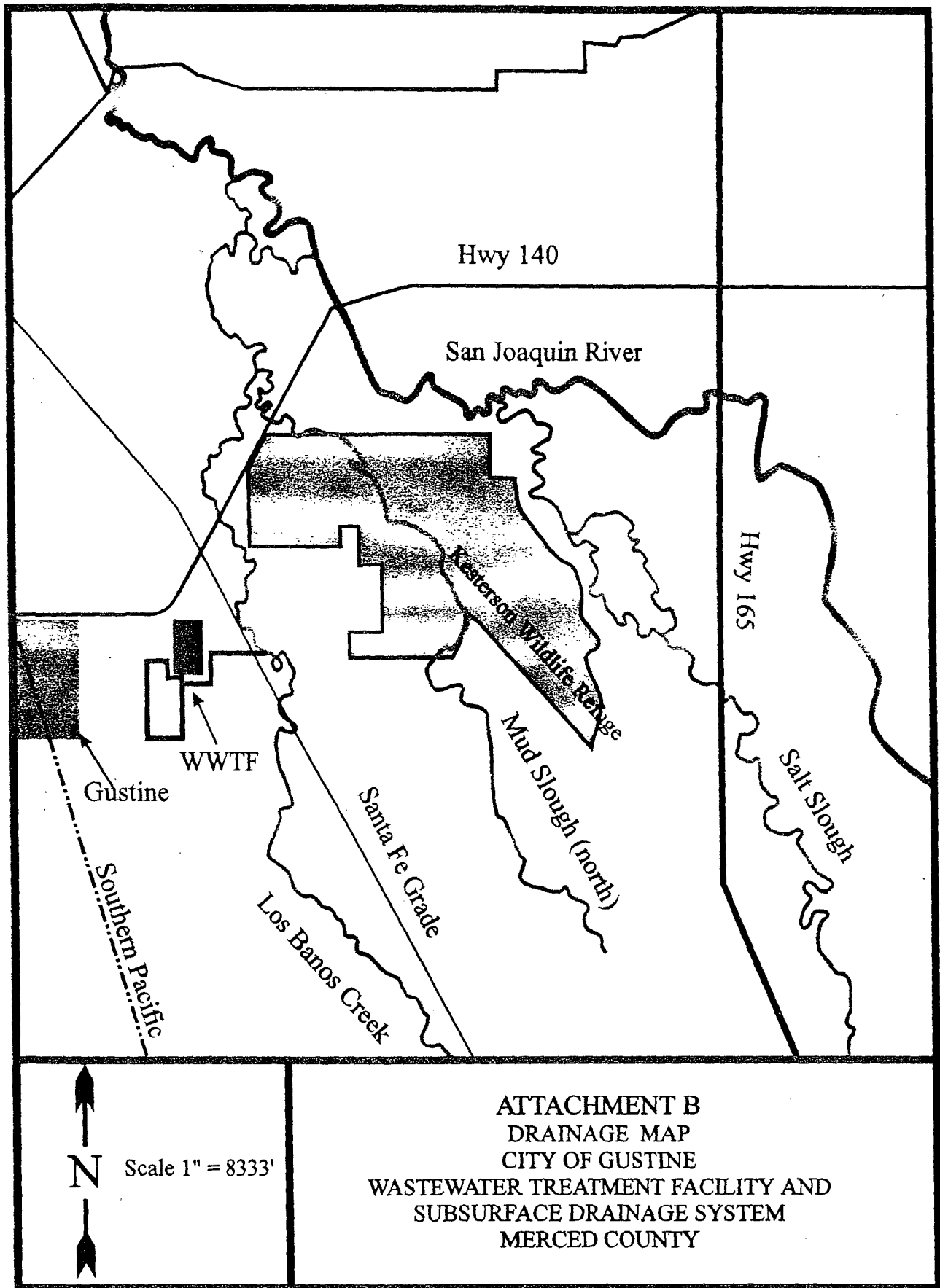
WDH:wdh/fmc: 2/27/98



Scale 1" = 2000'

**ATTACHMENT A
VICINITY MAP
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Sections 10, 11, and 15, T8S, R9E, MDB&M
Gustine and Ingomar, 7.5 Min. U.S.G.S Quad Maps



INFORMATION SHEET

CITY OF GUSTINE WASTEWATER TREATMENT FACILITY AND SUBSURFACE DRAINAGE SYSTEM MERCED COUNTY

The City of Gustine discharges about 1.0 mgd from its wastewater treatment facility into Los Banos Creek, a water of the United States and a tributary to Mud Slough (north) and the San Joaquin River (above the Merced River). Current discharges fail to meet the chronic and acute toxicity, pond dissolved oxygen, and receiving water limits. They also fail to meet total suspended solids and total coliform organism effluent limits consistently. The RWD and supplemental information indicates that about 55 percent of the flow and 70 percent of the WWTF organic loading is from Morningstar West and Beatrice Cheese Company, two milk processing facilities.

Staff calculations indicate, that from November 1996 through September of 1997, up to 90 to 97 percent of the organic loading might have been from industry. Staff calculated the monthly average daily BOD loading for each month by multiplying the monthly average flow in million gallons (MG) per day (mgd) by the monthly average influent BOD concentration in mg/l. The resulting value was then multiplied by the conversion 8.34 lbs/MG/mg/l. From November 1996 to September 1997 the monthly average BOD loading ranged from 10303 lbs/day and 30845 lbs/day and averaged 15500 lbs/day. The expected daily organic loading generated by 5190 to 5400 domestic users and based on a per capita wastewater generation rate of 60 gpd and a high strength BOD concentration of 400 mg/l would be 1039 to 1081 lbs/day. The Discharger projects that the quality of WWTF effluent, effluent discharged to the reuse area, and subsurface drainage water will be as follows:

<u>Constituent</u>	<u>WWTF Effluent</u>	<u>Applied Effluent</u>	<u>Subsurface Drainage</u>
BOD (mg/l)	45	40	< 1
TSS (mg/l)	90	60	< 2
Ammonia-N (mg/l)	< 15	< 5	< 0.5
Nitrate-N (mg/l)	< 1	< 1	< 1
TDS (mg/l)	1670	2309	5494
EC (µmhos/cm)	2778	3842	9140
Boron (mg/l)	0.9	1.24	2.96
Selenium (µg/l)	0.83	0.83	1.97

However, more recent data from the Discharger's self monitoring reports indicate that from May 1997 through September 1997 the monthly average EC of the effluent ranged from 5606 µmhos/cm to 7812 µmhos/cm. Daily values as high as 10000 µmhos/cm were recorded during May, July, and August of 1997. An effluent grab sample collected from the pond system on 22 November 1996 had sodium and chloride concentrations of 570 mg/l and 640 mg/l, respectively.

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The more recent data confirm that the wastewater influent is industry dominated and that effluent contains abnormal quantities of salt.

The Discharger completed a master plan in 1995 to upgrade and expand the capacity of its existing WWTF in phases from its current flow capacity of 1.2 mgd to 2.5 mgd to accommodate industrial and nonindustrial wastewater flows through the year 2014. This permit addresses the first phase of expansion to 1.4 mgd. The Discharger proposes to eliminate its current year-round discharge to Los Banos Creek and instead discharge year-round to 220 acres of adjacent grassland for pasture and nesting bird habitat.

Soils in the storage area are fine sandy, silty clays, and clayey silts to a depth of 21.5 feet below ground surface (bgs). The permeability of these soils is 2×10^{-7} to 4×10^{-7} cm/sec.

Local groundwater occurs in confined and unconfined aquifers. The unconfined aquifer is at 250 feet bgs, but there are shallow and perched groundwater bodies that may not be connected to the unconfined aquifer. Shallow groundwater beneath the proposed project area is 2 to 6 feet bgs and of moderate quality with concentrations of total dissolved solids of about 421 to 2515 mg/l, ECs of 658 to 2510 μ mhos/cm, and boron concentrations of 0.18 to 9.8 mg/l, and selenium concentrations from 0.7 to 3.6 μ g/l. Shallow groundwater is not currently used for municipal or domestic purposes. However, shallow groundwater is pumped into irrigation channels for groundwater level control, and in the channels it blends with irrigation water and water that supports wildlife habitat. Excess irrigation water eventually reaches Los Banos Creek. Source water for the City is provided by wells and is of good quality with concentrations of total dissolved solids (TDS) of about 680 to 840 mg/l, a EC of 1100 to 1300 μ mhos/cm, and nitrate concentrations of 29 to 41 mg/l. The average selenium concentration in the City's supply wells is about 0.8 μ g/l.

During the irrigation season (March through October), the Discharger proposes to pump effluent from the storage ponds to irrigate pasture and grassland habitat in the reuse area by the graded-border surface irrigation method. All irrigation tailwater will be recovered and returned to the storage ponds for reapplication. The Discharger also proposes to install a subsurface drainage system to control salinity levels in the root zone of the reuse area, maintain positive crop yield, and protect shallow groundwater quality. Further, from November through 15 March, it proposes to discharge up to 420 acre-feet annually at a maximum rate of 1.8 mgd of subsurface drainage water generated by the reuse area into a surface drainage channel immediately adjacent to and east of the proposed reuse area. The surface drainage channel discharges into Los Banos Creek approximately one channel mile (i.e., the actual length of the channel) east of the WWTF. Los Banos Creek then flows approximately 4.1 miles to Mud Slough (north), which flows approximately 3.5 miles to the San Joaquin River. Los Banos Creek is typically not used for irrigation purposes during the proposed period of discharge of subsurface drainage water.

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Wastewater discharge to the reuse area will overlap the nesting period for waterfowl and waterbirds, which is February through September. The reuse of effluent to establish and/or maintain nesting bird habitat could result in impacts to breeding and overwintering non-breeding waterbirds. It is unlikely that selenium related impacts will occur due to the low selenium concentration in the effluent. However, there could be impacts due to epidemic avian botulism and fowl cholera. As a result, a waterfowl and waterbird surveillance and control program is necessary to monitor for and mitigate potential impacts.

The Board amended the Basin Plan on 3 May 1996 to control agricultural subsurface drainage discharges. The amendment includes designation of beneficial uses, WQOs, and an implementation plan for achieving the WQOs for the Grassland Watershed water bodies. The Grassland Watershed water bodies include Mud Slough (north), Salt Slough, and water channels for the distribution of water supplies to the wetlands of the Grassland Watershed. The amendment also provides for the Board to adopt effluent limits in waste discharge requirements for discharges of subsurface drainage water from the Grassland Watershed to meet WQOs. For Mud Slough, the maximum and monthly average (15 March through 15 September) WQOs for boron are 5.8 and 2.0 mg/l, respectively; the maximum and monthly mean WQOs for molybdenum are 0.050 and 0.019 mg/l, respectively; and the 4-day average WQO for selenium, which becomes effective on 1 October 2010, is 5 µg/l. For Los Banos Creek, the daily maximum and monthly mean WQO concentrations for selenium are 20 µg/l and 2.0 µg/l, respectively.

The Basin Plan and its amendments prohibit certain activities associated with subsurface drainage water discharges. Specifically, Prohibition 6.a, states, in part:

“The discharge of agricultural subsurface drainage from the Grassland watershed to the San Joaquin River or its tributaries from ... any subsurface drain ... is prohibited, unless such discharge is governed by waste discharge requirements.”

Prohibition 6.e, states:

“Activities that increase the discharge of poor quality agricultural subsurface drainage are prohibited.”

The proposed discharge complies with Prohibition 6.a, as the discharge will be regulated by this Order. Prohibition 6.e does not apply to the Discharger's proposed subsurface drainage water discharge because the discharge will not result in an increase in the discharge of “poor quality” subsurface drainage. Prohibition 6.e originated as a policy (Policy IV) statement in the amendments to the Basin Plan that were approved by the Board on 8 December 1988. In the 1994 edition, this statement appears in both the policy section and the prohibition section. “Poor quality” is not defined in the Basin Plan, nor was it defined in the 1988, 1994, 1995, or 1996 amendments. The staff report supporting the 1988 Basin Plan amendments is the only document that describes “poor quality” for purposes of interpreting this particular policy. Pursuant to this

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description, activities that increase the discharge of "poor quality" agricultural subsurface drainage include the construction of new subsurface drainage facilities, or the expansion of existing facilities, in areas where the drainage already has an average selenium concentration of 20 µg/l, and where drainage already has an average selenium concentration greater than 5 µg/l unless certain conditions are met. It is, therefore, obvious that "poor quality" refers to discharges of subsurface drainage water containing selenium concentrations in excess of 5 µg/l. There is no indication that the Board, through subsequent amendments to the Basin Plan, intended to modify the meaning of "poor quality" in any way. The proposed discharge will be low in selenium with expected selenium concentrations below 5 µg/l.

The proceeding, however, does not obviate the need for additional measures to further reduce the salinity impacts that may result from the subsurface drainage water discharge. The anti-degradation analysis provided by the Discharger in its 6 March 1996 report indicates that the proposed discharge of subsurface drainage water into Los Banos Creek by itself could increase the EC of the creek to 3800 µmhos/cm when the flow rate and EC level in the creek are 10 cfs and the 3400 µmhos/cm, respectively. Staff analysis of the flow and EC data for the creek indicate that significant degradation, particularly under drought conditions, could also occur under the flow and EC conditions identified in the receiving water studies. Accordingly, both analyses show that there is reasonable potential for exceeding background EC and adversely affecting the beneficial uses and water quality in Los Banos Creek.

The Discharger proposes to mitigate this potential degradation of water quality in Los Banos Creek by (a) requiring its industrial users to develop and implement salinity control plans to reduce effluent EC concentrations, which in turn should significantly reduce the EC of the subsurface drainage water; and/or (b) discharging the subsurface drainage water into Los Banos Creek when flows in Los Banos Creek are at least 14.5 cfs. However, even when flows are 14.5 cfs, the discharge could increase the EC levels in the creek from 710 to about 2050 µmhos/cm.

Subsequent to the studies and analyses conducted by the Discharger, the U.S. Bureau of Reclamation (Bureau) and drainage dischargers in the Grassland Subarea (Bureau designation) implemented the Grassland Bypass Project. The project utilizes a 28-mile segment of the San Luis Drain (Grassland Bypass) to convey agricultural drainage waters generated in the Grassland Subarea to the San Joaquin River via a six mile segment of Mud Slough. As a result, drainage waters that were previously conveyed to the San Joaquin River via Mud and Salt Sloughs are now concentrated in Mud Slough.

Mud Slough is listed in the Basin Plan as a Water Quality Limited Segment for selenium, TDS, pesticides, and toxicity. The Basin Plan also lists the San Joaquin River from Sack Dam to Vernalis as a Water Quality Limited Segment for selenium, toxicity, diazinon, parathion, carbaryl, eptam, Group A, toxaphene, DDT and endosulfan. Group A refers to the following

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chemicals, either individually or in combination: aldrin, dieldrin, endrin, heptachlor, heptachlor epoxide, chlordane (total), hexachlorocyclohexane (total), endosulfan (total) and toxaphene. The Basin Plan also designates numerical WQOs for salinity in the San Joaquin River at Vernalis. The 30-day running average EC objective from April to August is 700 $\mu\text{mhos/cm}$. The 30-day running average EC objective from September to March is 1000 $\mu\text{mhos/cm}$. From October of 1985 to September of 1996, the April to August EC limit was violated 58% of the time and the September to March salinity limit was violated 13% of the time. The Discharger will be replacing its direct discharge and salt load to Los Banos Creek with a subsurface drainage discharge to Los Banos Creek. The Discharger will only discharge from the subsurface drainage system when it needs to remove salts from its reuse area to maintain crop yields. The subsurface drainage discharge is expected to occur every two to three years. While the overall annual salt load to Los Banos Creek and its downstream waterways is expected to be less than currently discharged, discharges from the subsurface drainage system may be more concentrated when they occur. The concentrated salt loading resulting from the City of Gustine's proposed discharges of subsurface drainage water have the potential to contribute to the violations of the water quality objectives in Mud Slough and the San Joaquin River at Vernalis.

Salinity controls on the discharge, such as influent EC reductions and timing restrictions for the proposed subsurface drainage discharges and/or application of low EC reclaimed water at evapotranspiration rates, are necessary to reasonably control Gustine's contribution to adverse impacts on water quality in Los Banos Creek, Mud Slough, and the San Joaquin River, and particularly salinity objective violations now occurring in the San Joaquin River at Vernalis. The effluent EC goal of 1,800 $\mu\text{mhos/cm}$ in this Order is based on the Discharger adding 500 $\mu\text{mhos/cm}$ to the upper end of its groundwater supply EC of 1300 $\mu\text{mhos/cm}$. Small increases in the salinity of groundwater underlying treatment and storage ponds and the reuse area may result from the discharge. The Antidegradation Analysis conducted by the Discharger indicates that potential changes in the reuse areas will be controlled by the subsurface drainage system and will be insignificant. Groundwater degradation caused by the ponds will be minor, limited to POCs, and are acceptable with implementation of the described pest practicable technology and need for wastewater utility service to accommodate housing and economic expansion in the area, a benefit to the people of the State.

The Discharger needs to establish an adequate pretreatment program to regulate the EC and TDS of industrial discharges into its collection system. Therefore, this Order requires the Discharger to implement a pretreatment program similar to that required for NPDES discharges. The program will require the Discharger to establish local limits for EC and total dissolved solids for its industrial dischargers.

Pursuant to Title 22, CCR, Section 60323, the Discharger's consultant submitted a water reuse engineering report dated 27 August 1997 to DHS. The DHS approved the report by letter dated 8 September 1997. The Board has consulted with DHS regarding the Discharger's discharges for reuse purposes and incorporated DHS recommendations into this Order.

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The City of Gustine approved a negative declaration for proposed project in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The Board has reviewed the negative declaration and compliance with these waste discharge requirements will mitigate or avoid potential adverse water quality and resources impacts.

WDH:wdh/fmc:2/27/98

